



Intro to Virtual Reality

V5.8



Unreal Training

Intro to Virtual Reality

Objectives

- Discuss VR basics and best practices
- Utilize the VR Template to work across a range of VR devices.
- Utilize the Grabbable component to interact with objects in your scene
- Understand the Enhanced Input system
- Explore potential modifications to the VR Pawn and controller menu
- Modify Spectator Cam and Menu to smoothly follow HMD view

Agenda

- What is VR?
- Virtual Reality Best Practices
- What is Open XR?
- Collab Viewer Template
- VR Template
- VR Set-up
- VR Spectator
- Interactivity
- Package for Mobile or PC VR
- Tips and Tricks!

GOAL



Images courtesy of Cameron Swanson

Intro to Virtual Reality

What is **VR**?

"It's like you're there!"

What is VR?

- Virtual Reality is stereoscopic
 - Renders a separate view for each eye
 - Creates depth perception
 - Simulates how our own eyes perceive the world
- Immersive, scale perception
- Can be very interactive



What is VR?

AR is *not* VR

- AR stands for Augmented Reality
- Sometimes referred to as “Mixed Reality”
- Combines elements from the real world with virtual elements
- VR is typically 100% virtual

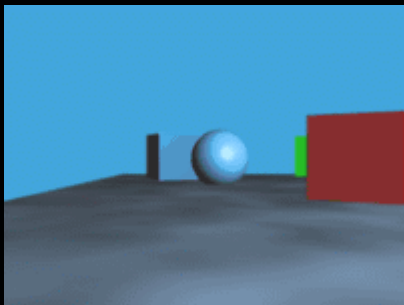
What is VR?

360 video is *not* VR

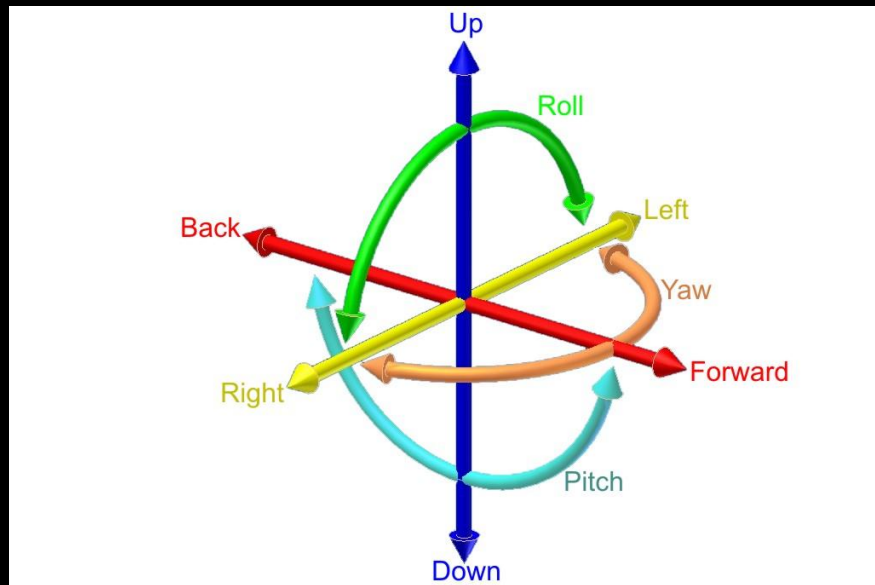
- Often confused to be the same thing, but some similarities
- Sometimes uses VR headsets to view 360 video
- Not interactive
- Can be stereo
- Only 3 DoF (degrees of freedom)

Degrees of Freedom

- 3 DoF
- 6 DoF
- Parallax



Credit: Nathaniel Domek



Limitations

- Limited FOV
 - No zoom
 - Limited screen space effects
 - Expensive to render
 - Resolution
-
- Some people just don't want to wear headsets!



Health warnings

- Motion sickness
- Dizziness
- Eye strain
- Neck strain
- Headaches
- Headset hygiene



Images courtesy of Cameron Swanson

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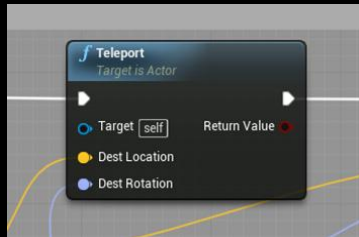
Virtual Reality Best Practices

Motion Sickness

- Why?
 - Your inner ear doesn't feel what your eye is seeing
- What things cause it?
 - Low FPS
 - Taking control of the camera
 - Fast movement or sudden change in direction
- How to prevent it?
 - Ground the player
 - Move the world, not the camera
 - Teleport
 - Some mechanical hardware can counteract this

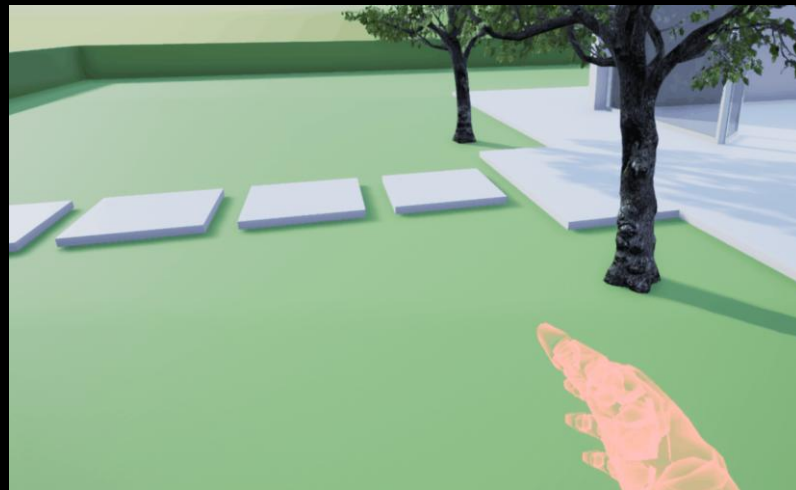
Movement

- Teleport
- Fade to black
- Slow and constant movement
- Avoid overwriting rotational input
 - Move in straight lines
 - Roller coasters are a good way to make your player want to vomit
- Don't take control of the camera
 - No swooping or panning cinematic cameras



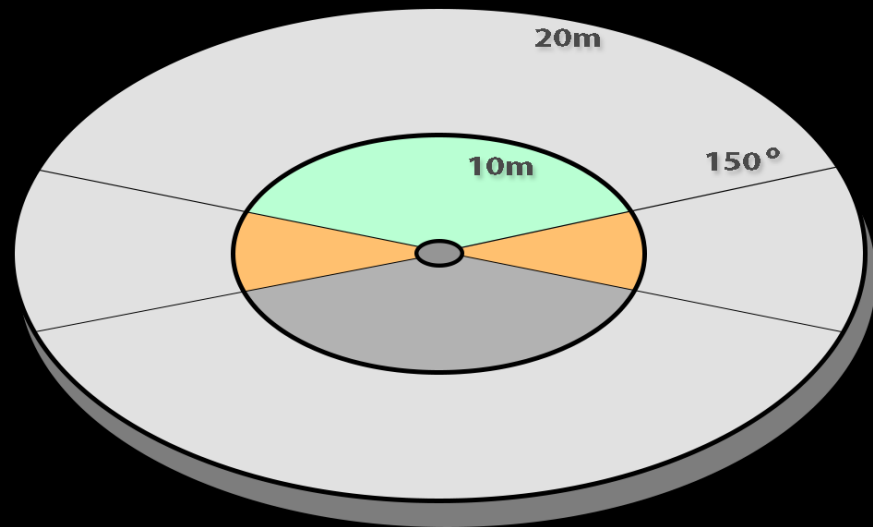
Teleporting

- Ships default both with CollabViewer and VR Template
- Handled on the Pawn
- Quick and comfortable
- Not 'realistic' movement
- Requires NavMesh
- Required Collision



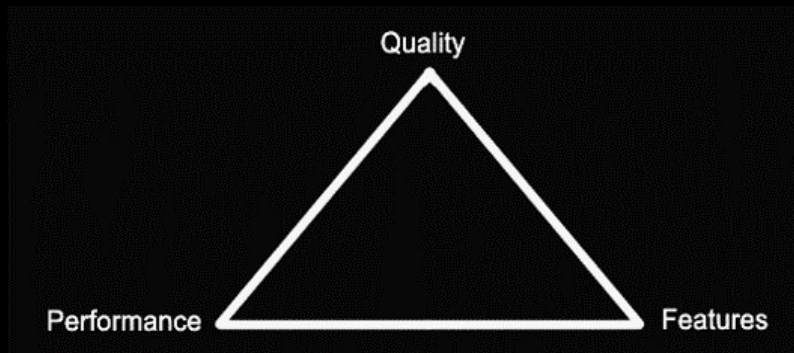
Perceived Distances

- Hard to see meaningful parallax beyond 20 meters
- Beyond 20m consider using cards for background assets
- Comfort range is up to 0.5 to 10 meters
- Comfort stationary angle ~150 degrees



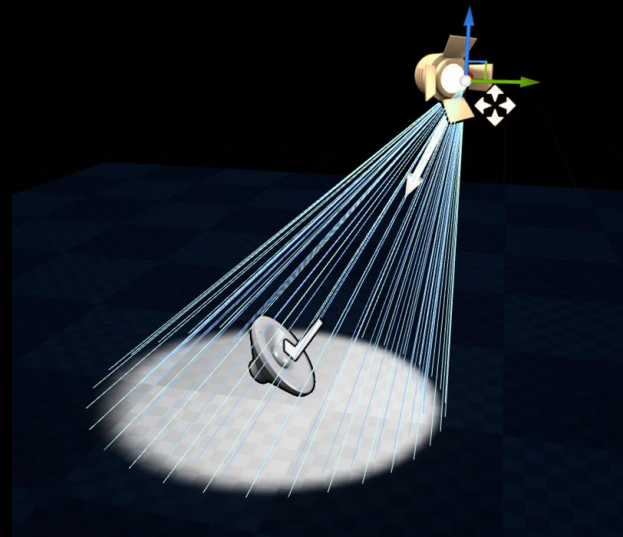
Performance

- Most desktop VR should run at 90 fps
- Mobile is usually 72 fps (but can be as high as 120)



Focusing on points of interest

- Getting users to look when you want them to is hard!
- Use audio cues
- Use visual cues
- Users will do their own thing regardless
- Newcomers to VR treat it like TV
- Tactility is an immersion multiplier



Intro to Virtual Reality

No hiding!

- Users have the freedom to look wherever they want
- Environments need to be fully built out
- No hiding anything!



VR - Best Practices

“More like guidelines than actual rules...”

Images courtesy of Cameron Swanson

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What is OpenXR?

"One plug-in to rule them all"

OpenXR

Currently in Unreal Engine supported by:

- Windows Mixed Reality
- HoloLens
- Meta
- SteamVR (Vive, Valve Index, Varjo, etc)

OpenXR is a royalty-free, open standard that provides high-performance access to Augmented Reality (AR) and Virtual Reality (VR)—collectively known as XR—platforms and devices.

- API for XR applications
- The interface between an application and runtime
- Mechanism for interacting with VR/AR/MR systems in a platform-agnostic way

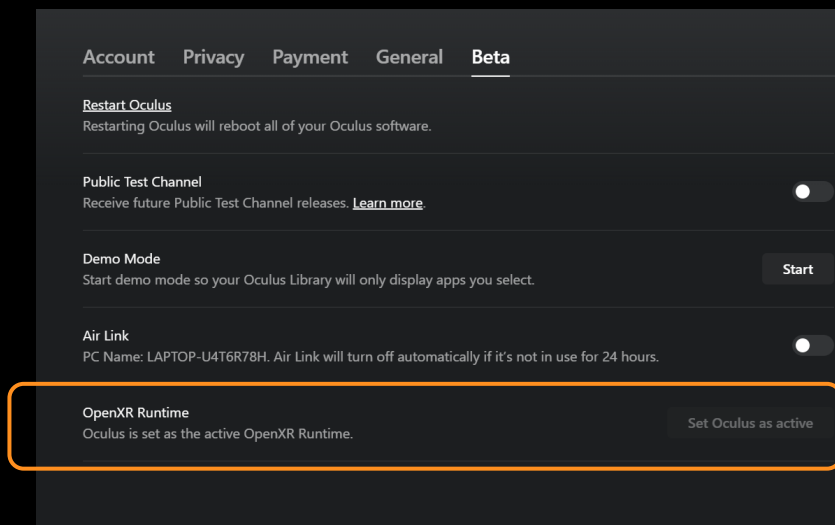
<https://www.khronos.org/openxr/>

<https://www.khronos.org/registry/OpenXR/specs/1.0/html/xrspec.html>

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Enable OpenXR for SteamVR, WMR, or Oculus

... or you'll have unexpected behavior.



Account Privacy Payment General **Beta**

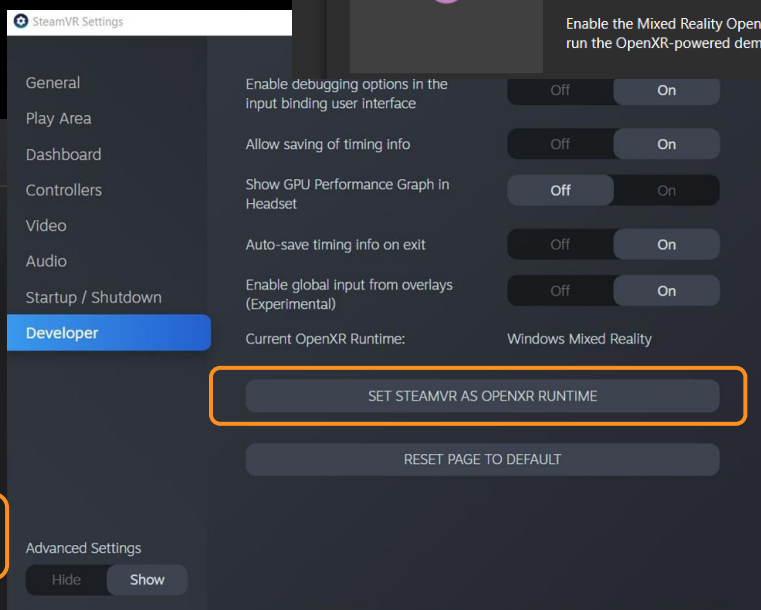
Restart Oculus
Restarting Oculus will reboot all of your Oculus software.

Public Test Channel
Receive future Public Test Channel releases. [Learn more](#)

Demo Mode
Start demo mode so your Oculus Library will only display apps you select. **Start**

Air Link
PC Name: LAPTOP-U4T6R78H. Air Link will turn off automatically if it's not in use for 24 hours.

OpenXR Runtime
Oculus is set as the active OpenXR Runtime. **Set Oculus as active**



SteamVR Settings

- General
- Play Area
- Dashboard
- Controllers
- Video
- Audio
- Startup / Shutdown
- Developer**

Enable debugging options in the input binding user interface: Off On

Allow saving of timing info: Off On

Show GPU Performance Graph in Headset: Off On

Auto-save timing info on exit: Off On

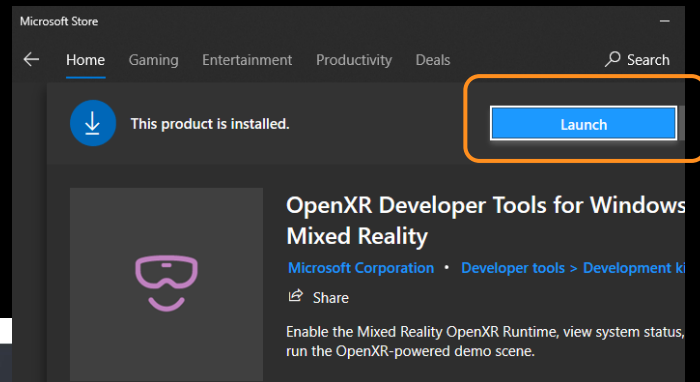
Enable global input from overlays (Experimental): Off On

Current OpenXR Runtime: Windows Mixed Reality

SET STEAMVR AS OPENXR RUNTIME

RESET PAGE TO DEFAULT

Advanced Settings
Hide Show



Microsoft Store

Home Gaming Entertainment Productivity Deals Search

Download This product is installed. **Launch**

OpenXR Developer Tools for Windows Mixed Reality
Microsoft Corporation • Developer tools > Development kit

Share

Enable the Mixed Reality OpenXR Runtime, view system status, run the OpenXR-powered demo scene.

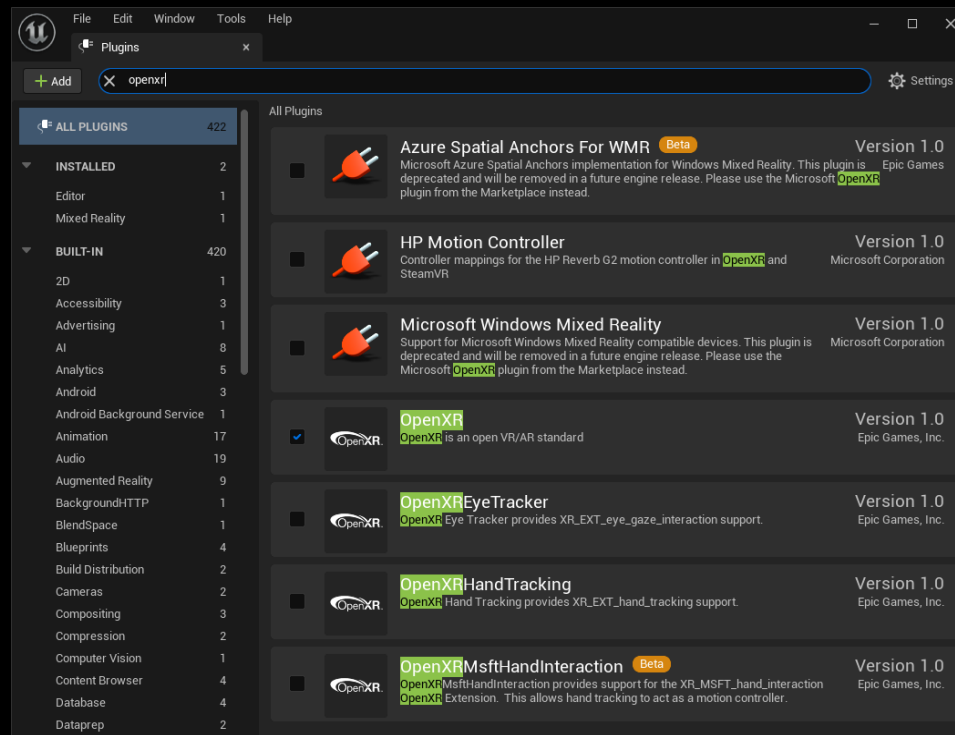
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Plugins for OpenXR

“OpenXR” - for general support. The only one you need enabled.

Note: other plugins seen here are for hand and eye tracking.

Additional OpenXR plug-ins may also be found on the Marketplace.



Images courtesy of Cameron Swanson

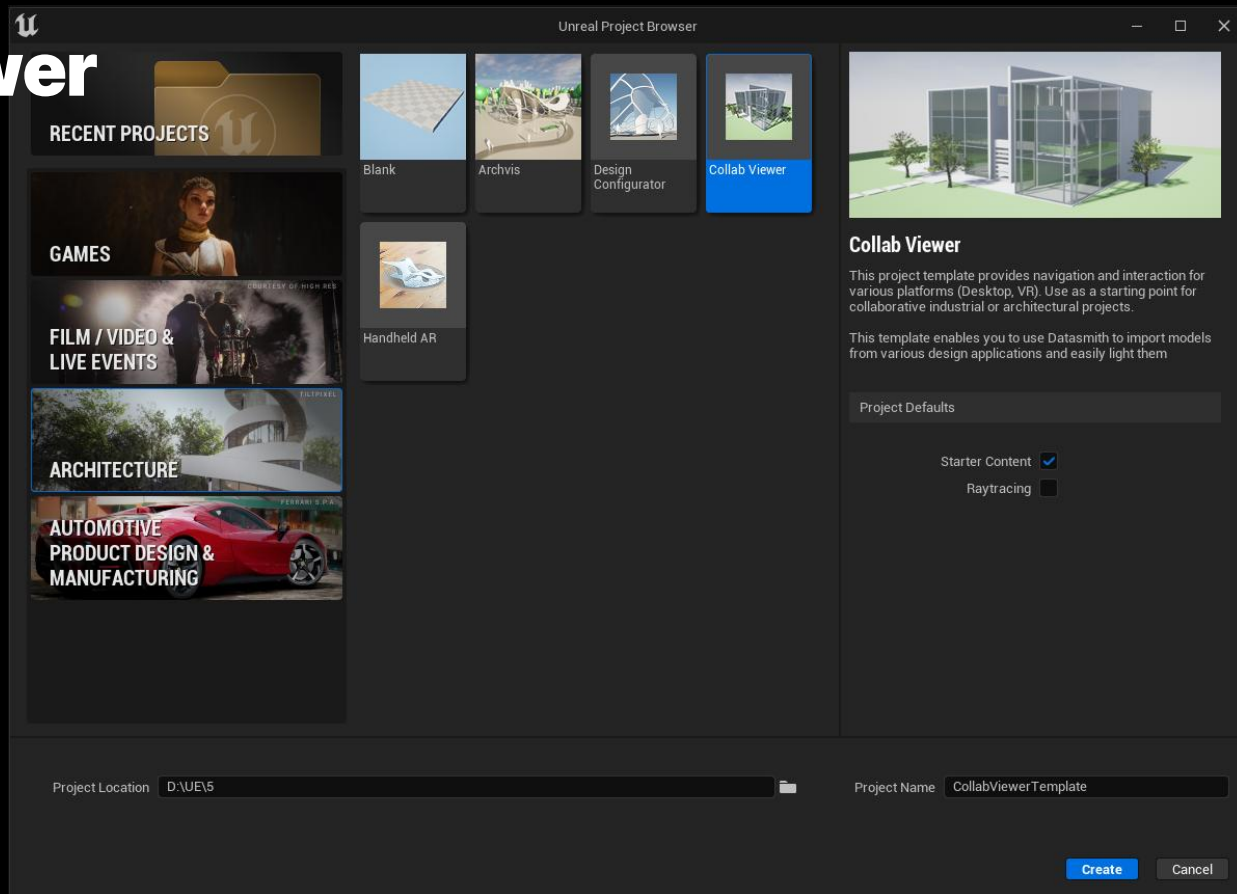
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Collab Viewer Template

A brief fly-by

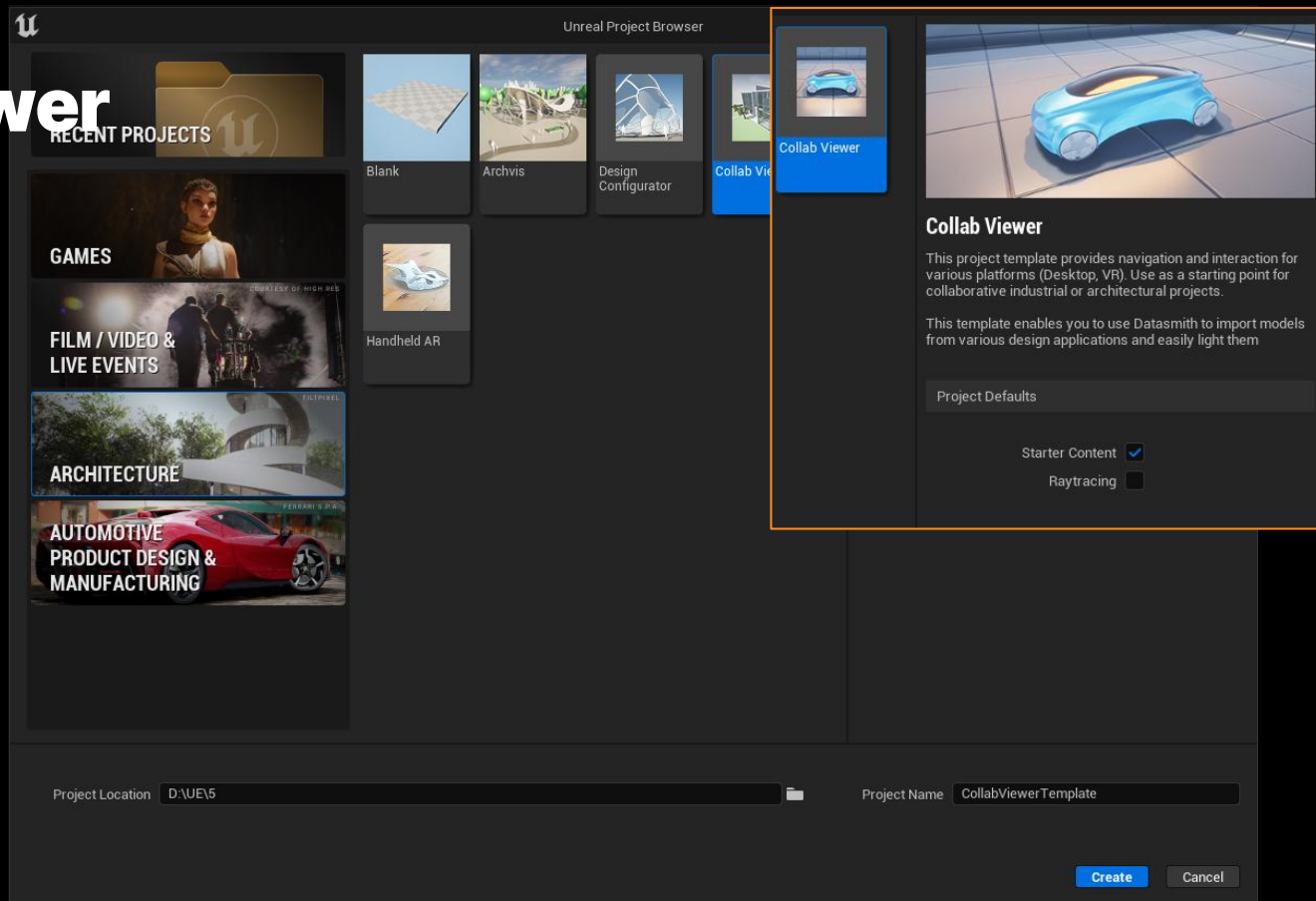
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CollabViewer



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CollabViewer



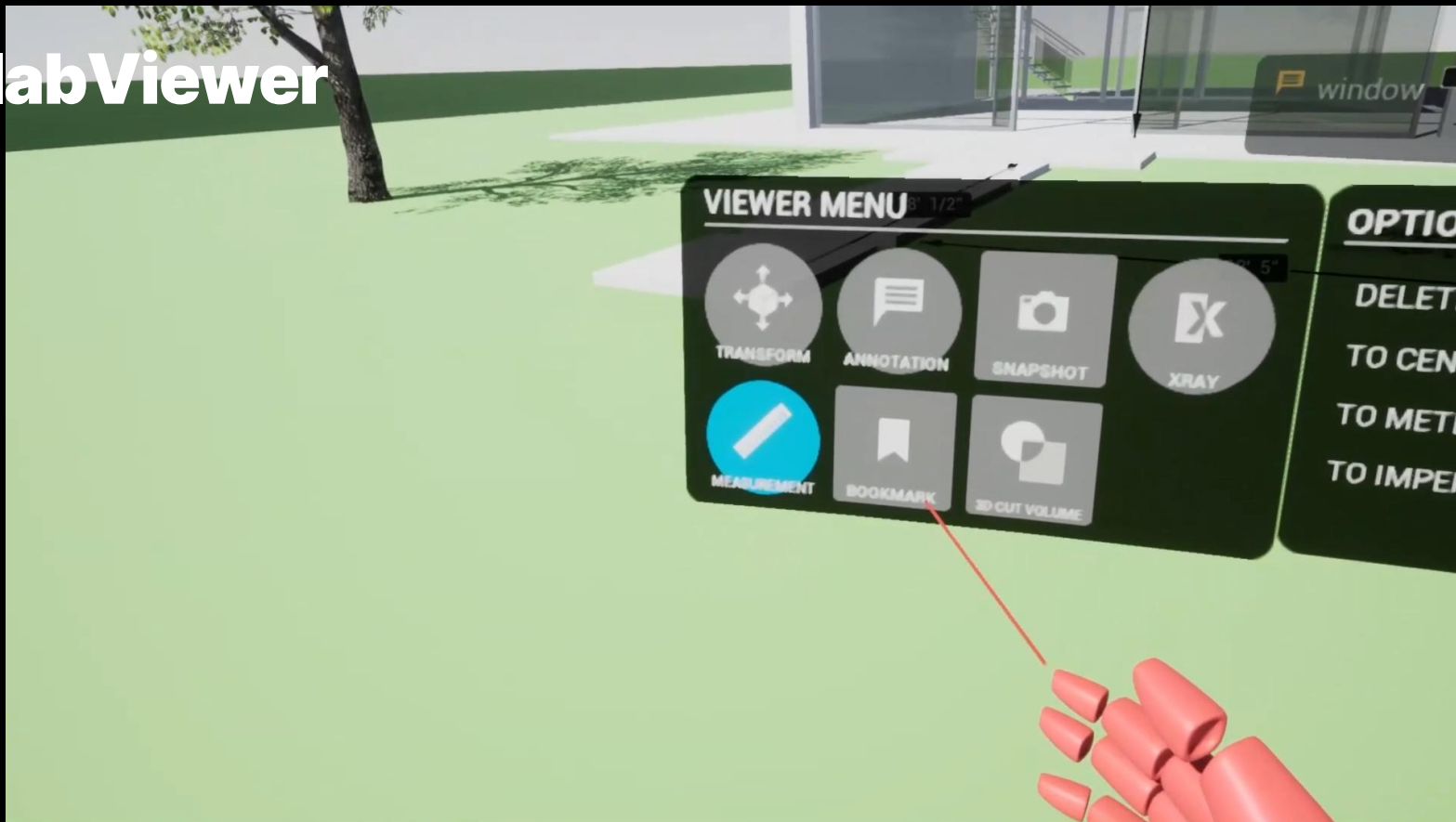
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CollabViewer



Simon Blakeney's CollabViewer Lab at Unreal Fest 2022

CollabViewer



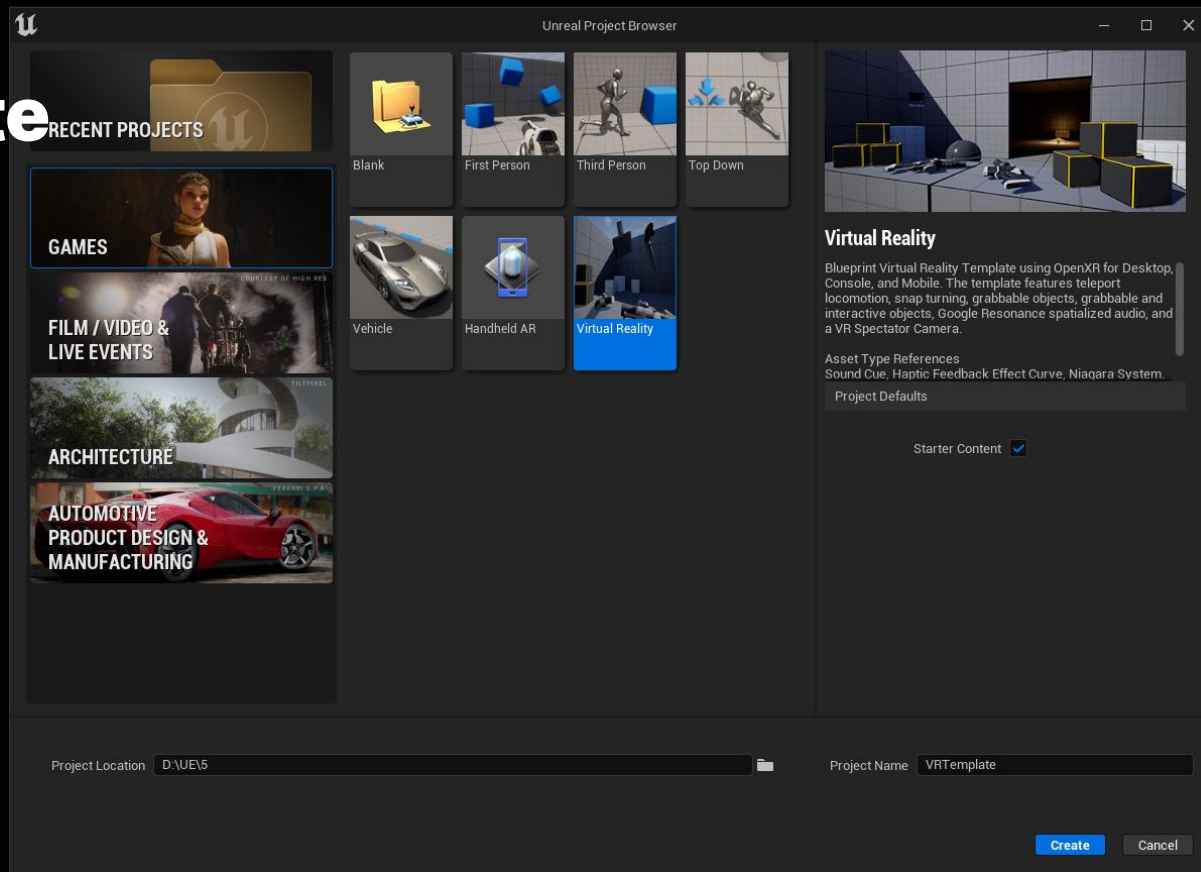
Images courtesy of Cameron Swanson

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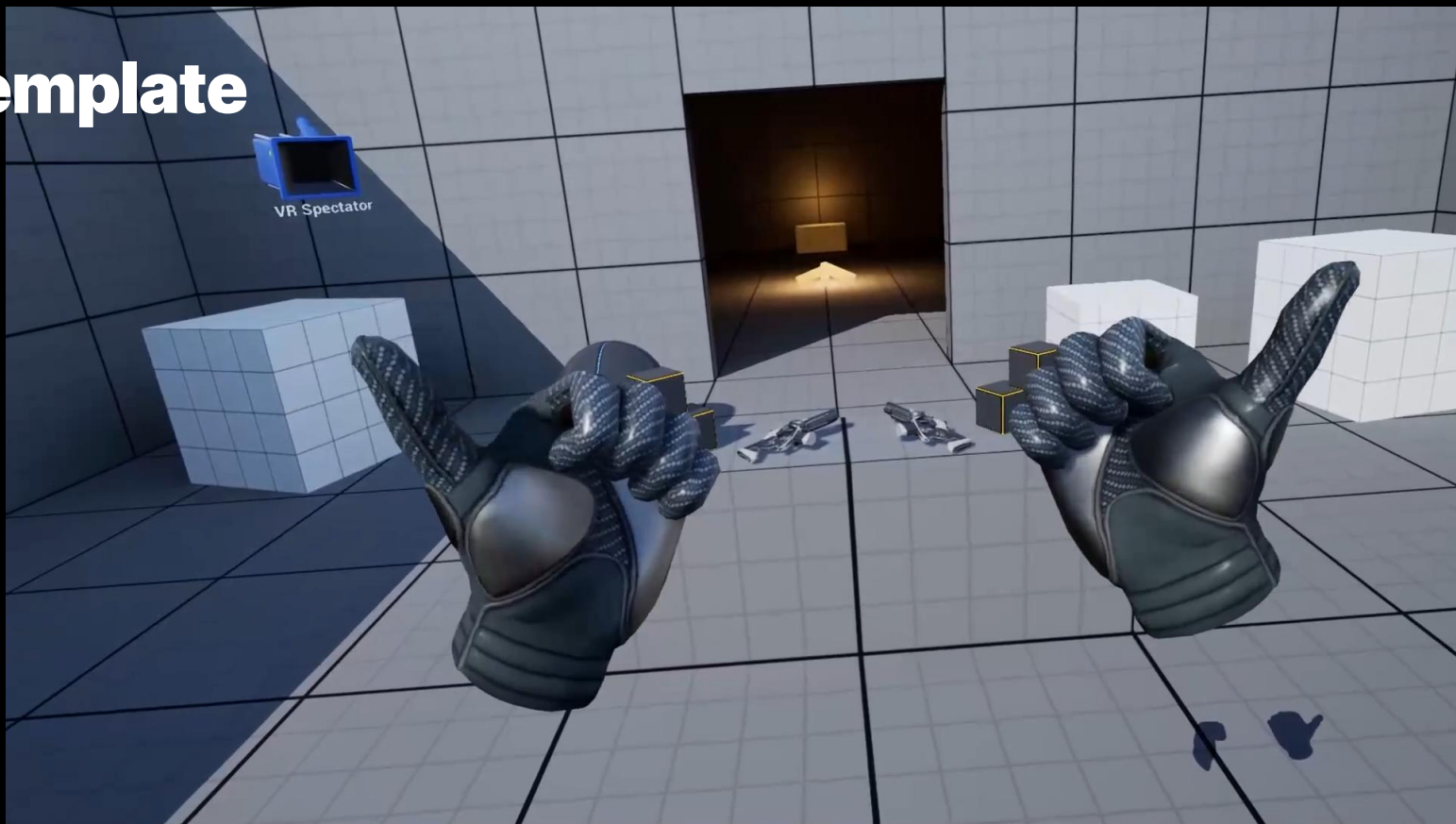
VR Template

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VR Template



VR Template



Build to Android and Windows

Works out of the box!

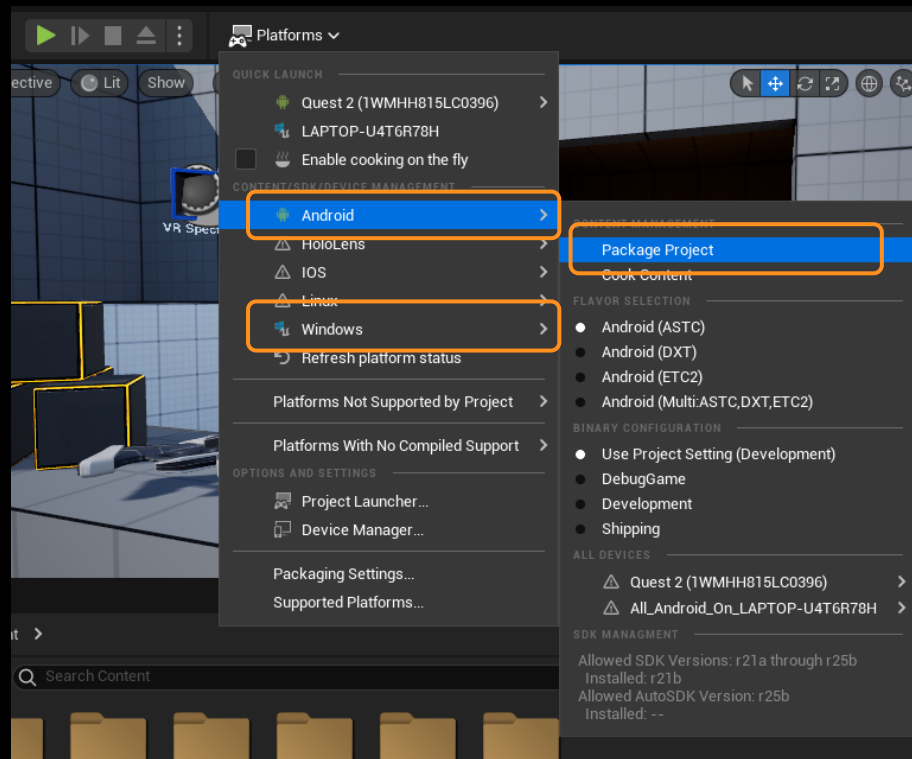
Just make sure you've set up Android.

<https://docs.unrealengine.com/SharingAndReleasing/Mobile/Android/GettingStarted/>

<https://github.com/NuLL3rr0r/unreal-android-installer>

Community guide for Unreal Engine 5.7 for Meta devices:

<https://dev.epicgames.com/community/learning/tutorials/d60x/unreal-engine-5-7-x-for-meta-quest-vr>



Images courtesy of Cameron Swanson

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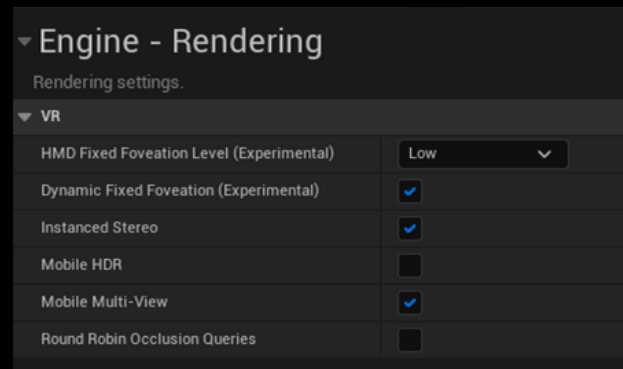
VR Set-up in 5.8

How Does Virtual Reality Work in UE5?

- Variety of Headsets
 - Meta Rift/Quest, HTC Vive/Focus, Windows Mixed Reality, Valve Index
- Desktop vs Mobile
 - Portability vs Fidelity
 - Pixel streaming, NVIDIA CloudXR, Air Link, Virtual Desktop, and Vive Business Streaming allow wireless Desktop VR
- OpenXR
 - One-stop shop for input (even better with new Enhanced Input!) Built into VR Template and Collab View Template
- Lumen / Nanite / DX12 – Available but not yet stable

VR – Basic Project Settings

- Mobile HDR allows for Post Processing in a mobile headset (e.g. Quest/Pico/Focus) but is extremely expensive.
- Round Robin Occlusion Improves VR by alternating rendering occlusion between one eye each frame instead of both at the cost of 1 frame of latency.

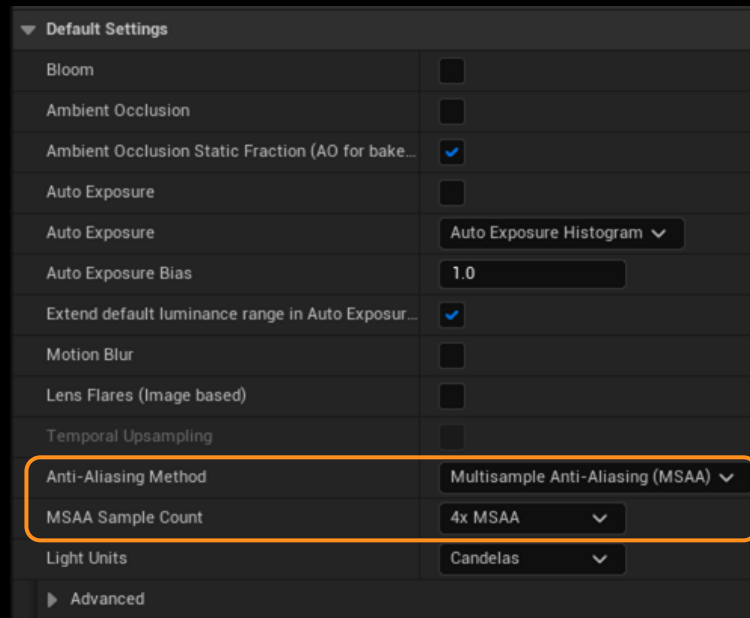
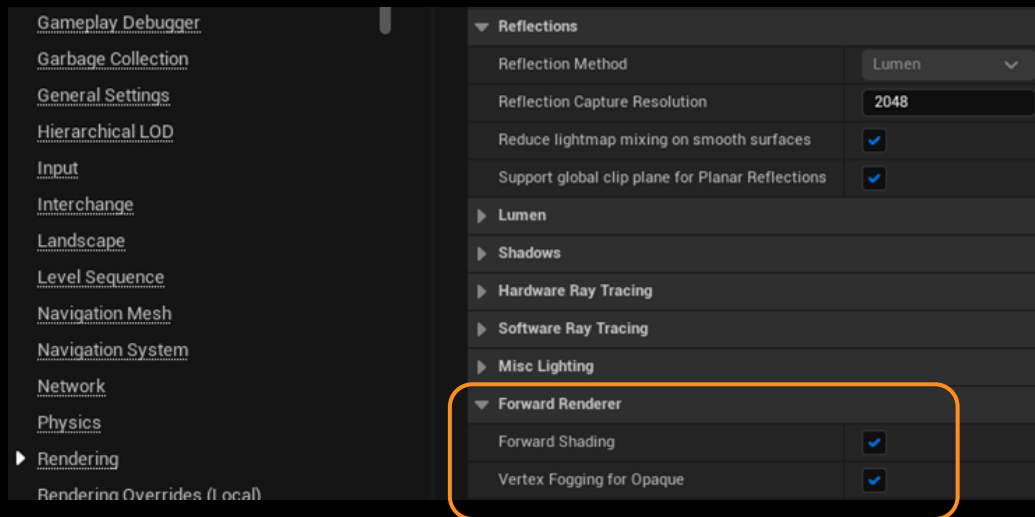


- To provide an optimized path for Stereo Rendering on CPU, use Instanced Stereo (desktop) and Mobile Multi-View (mobile).

VR - Forward Rendering

Improved performance over Deferred Rendering if most lighting is baked. Also allows for MSAA:

- Less blur than TAA
- Can produce more 'jaggies' than TAA

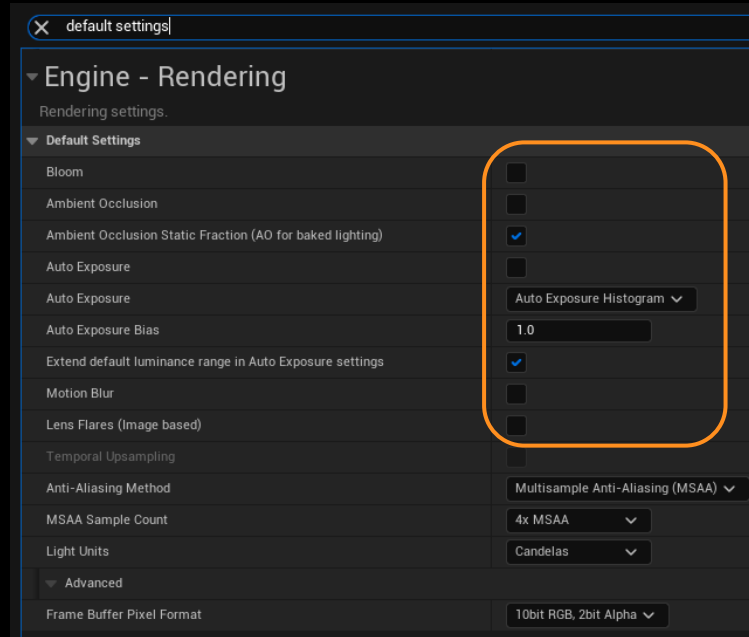


VR – Post Processing

Expensive!

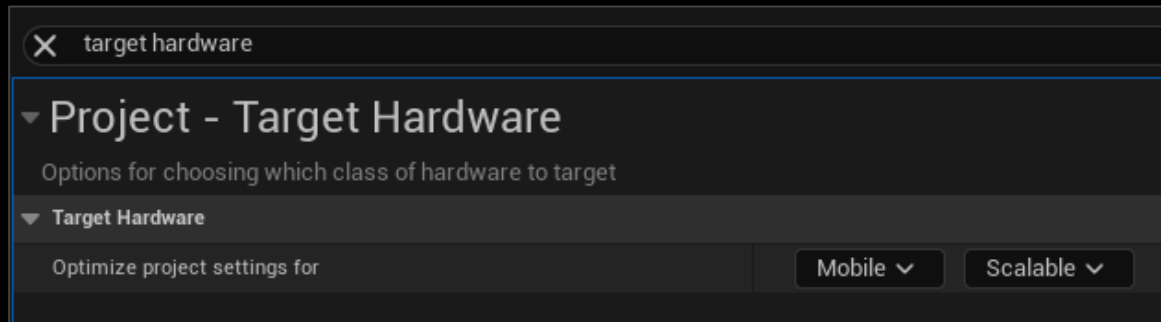
Disable most at engine level, turn on as desired in Post Processing Volume

- Set SSAO low or off
- Set bloom low or off completely
 - Convolution bloom looks great but is very expensive
- Set SSR off in PPV
 - Look incorrect in Deferred for VR
 - Unsupported by Forward
- Turn off Lens Flares
- Turn off Motion Blur
- Avoid screen effects (light rays, lens dirt, film grain)



VR - Target Hardware

A quick way to adjust many project settings to work better in VR is to change your Target Hardware to Mobile / Tablet and Scalable 3D or 2D.

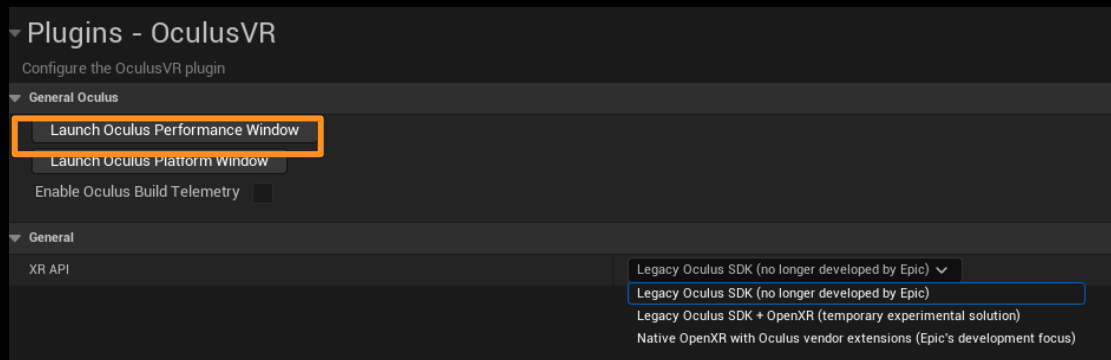


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VR - Target Hardware

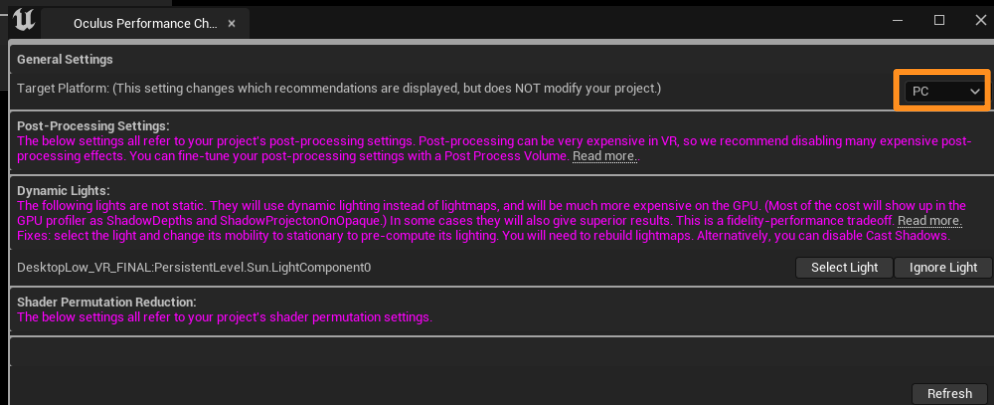
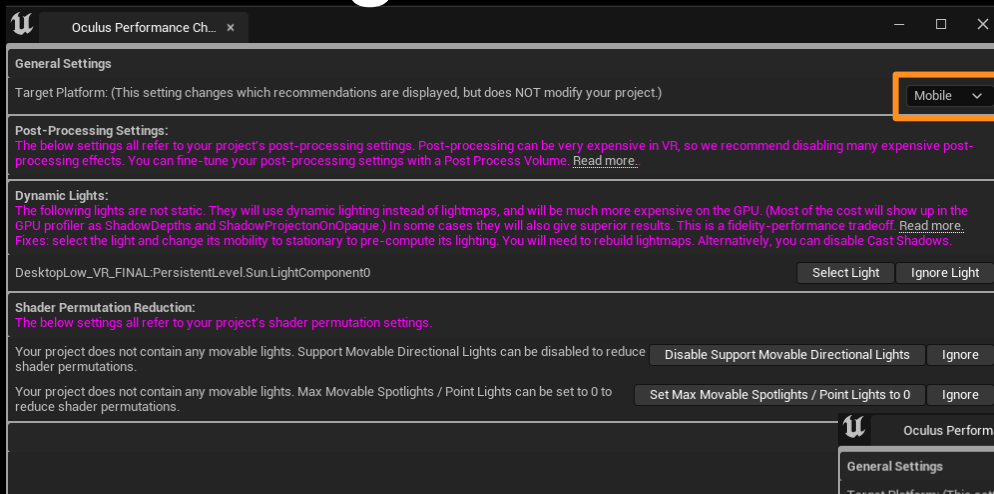
Use the **Oculus Performance Check** to flag anything that might reduce VR performance for PC or Mobile VR, even if not using an Oculus device.

(OculusVR plugin)



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VR - Target Hardware



Images courtesy of Cameron Swanson

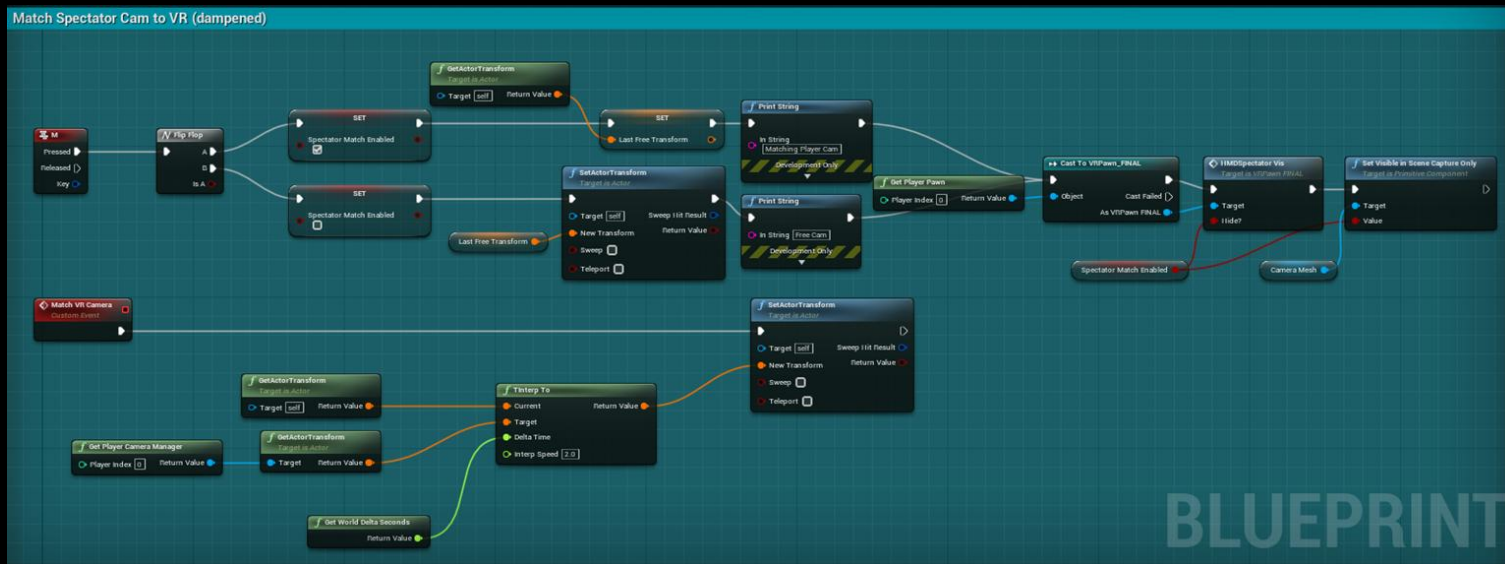
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VR Spectator

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VR Spectator

Dampened Camera is more comfortable for viewing audience.



Images courtesy of Cameron Swanson

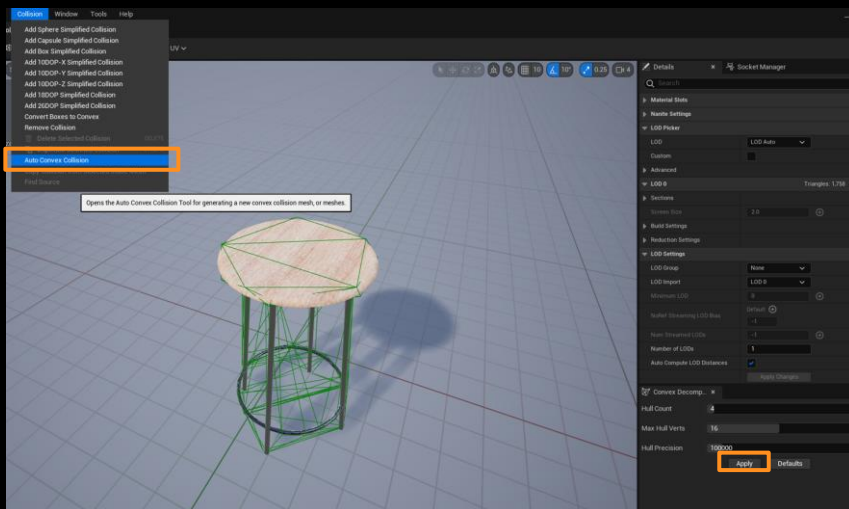
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Interactivity

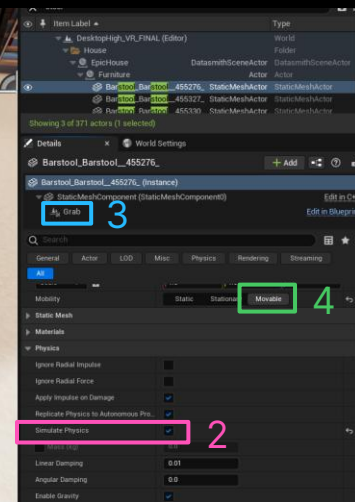
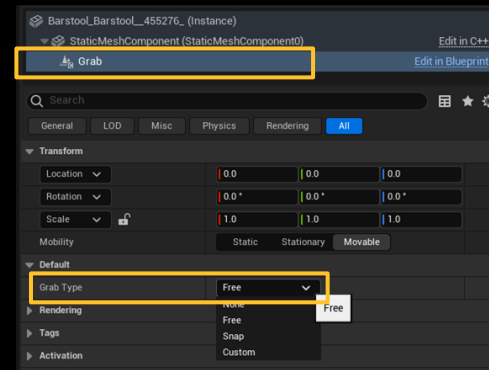
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Create New Grabbable Object

1. Give **Static Mesh** a collision volume
2. Set **Simulate Physics**
3. Add **GrabComponent**
4. Set **Movable**
5. Set **Grab type**



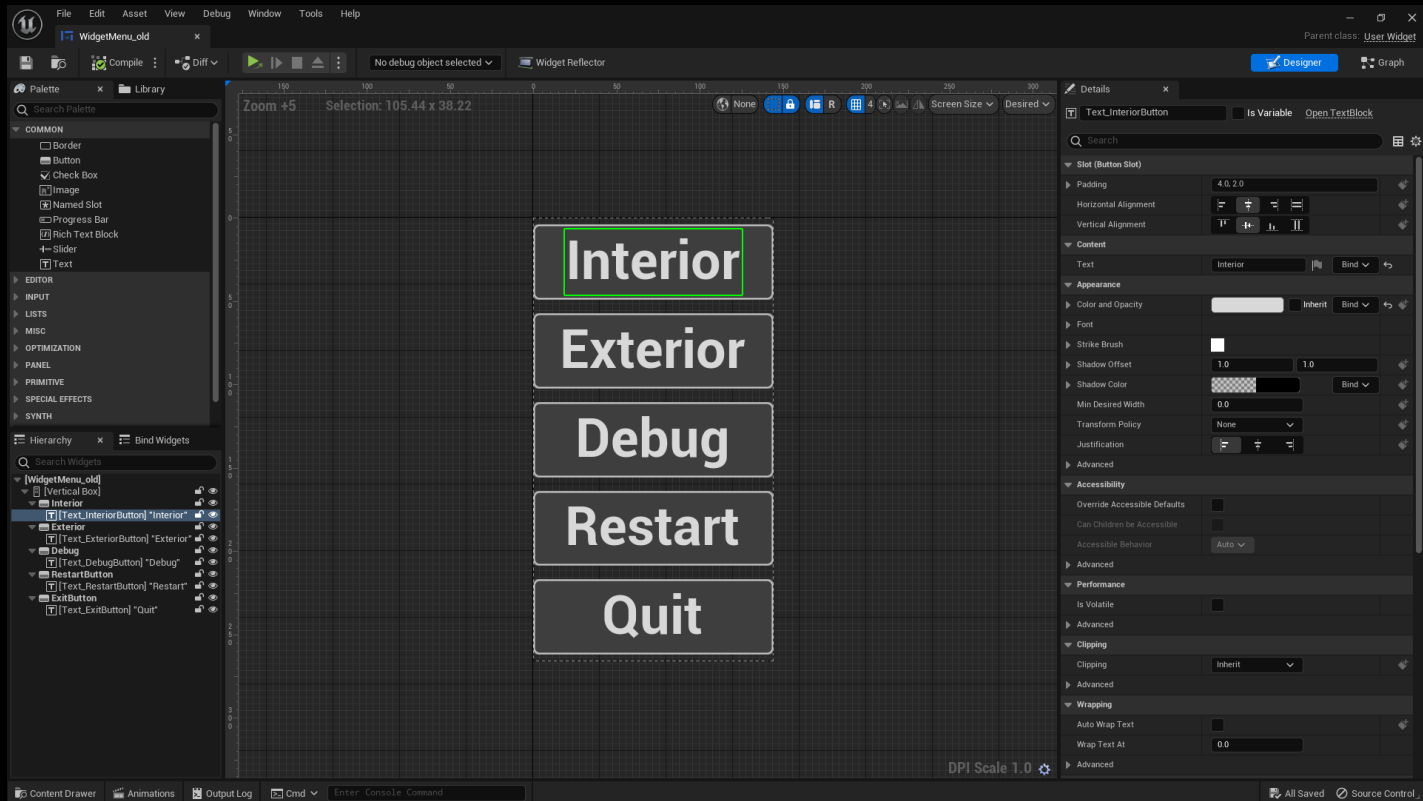
5



VR Interaction - Grabbing

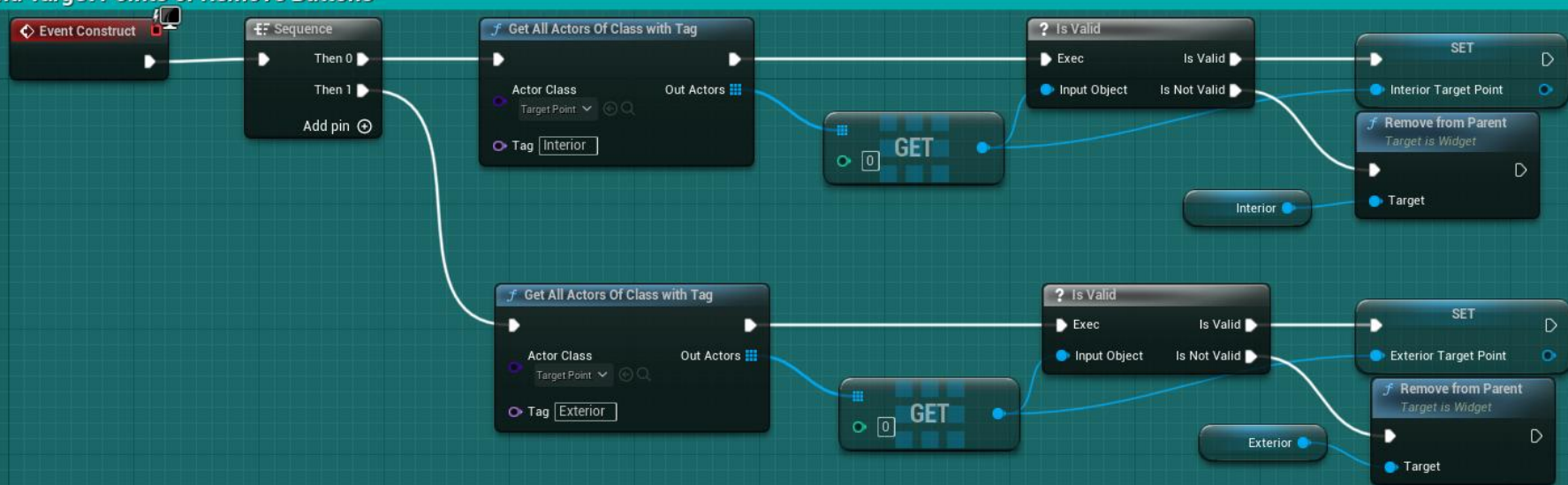


Adding Menu Items



Adding Menu Items

Find Target Points or Remove Buttons



Adding Menu Items

Interior Button - Teleport to Target Point with Interior Tag



Exterior Button - Teleport to Target Point with Exterior Tag



DebugButton - Pull Up Stats



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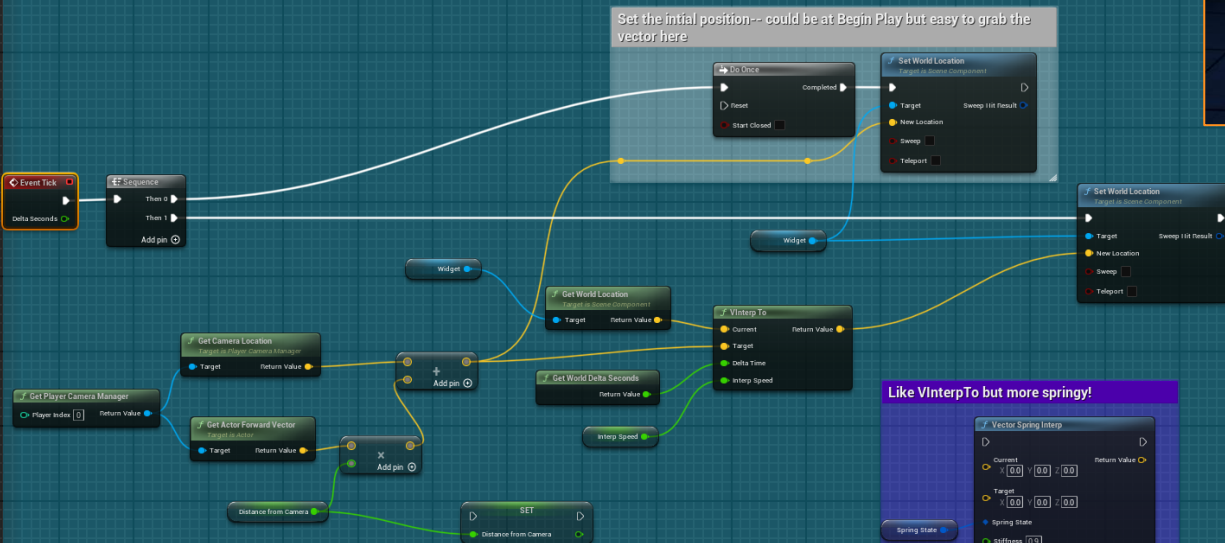
Changing the Menu

Menu_Ex > Event Graph

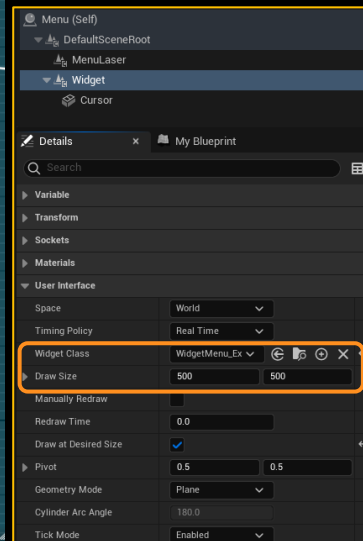
Move the widget to a comfortable location *in front of camera*

Move the widget to a comfortable location *in front of camera*

Set the initial position-- could be at Begin Play but easy to grab the vector here



Like VinterpTo but more springy!



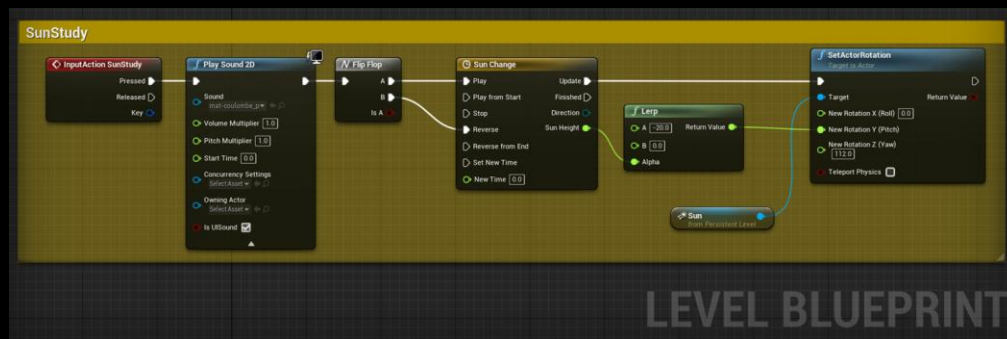
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Sun Study

Expensive but beautiful!

If can't afford Moveable sunlight, use
Stationary and turn down Intensity.

(Note Low vs High Maps)



Images courtesy of Cameron Swanson

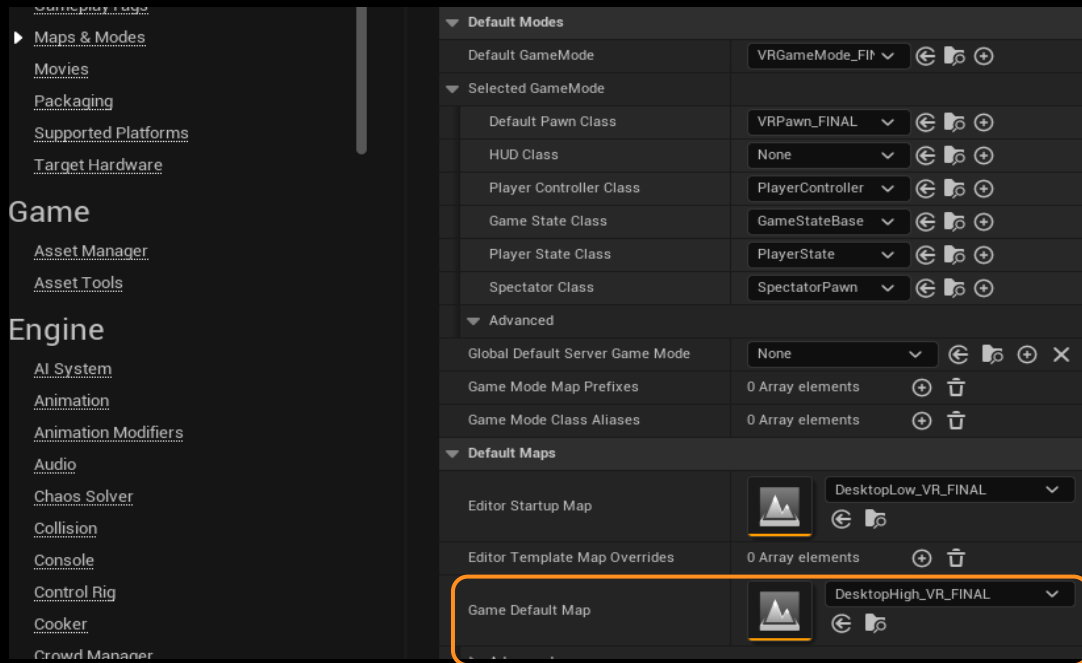
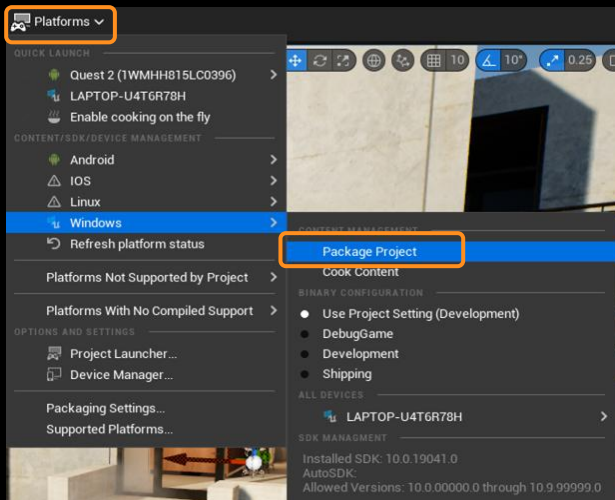
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Package for Mobile or PC VR

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PC - Package an .exe file

Ensure the Game Default Map is set to match the specs of the target PC.

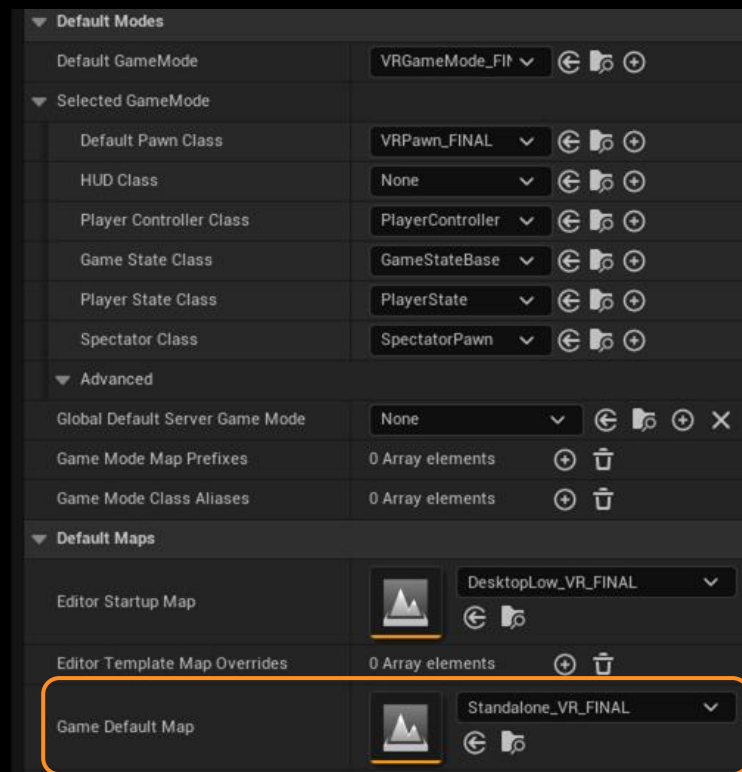
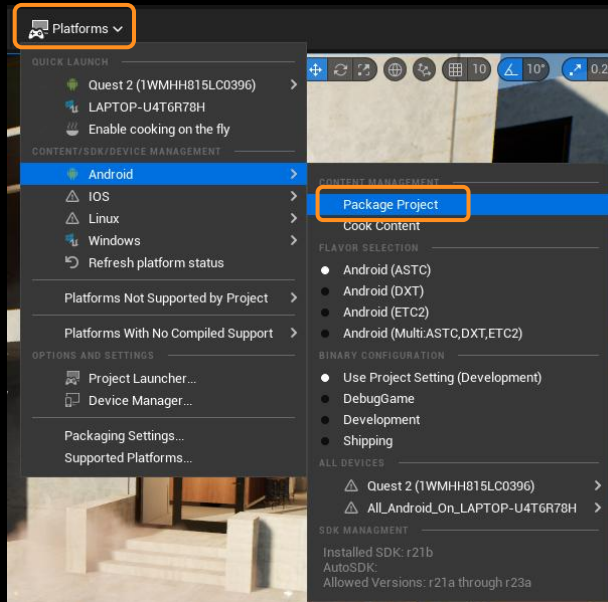


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Mobile - Package an .apk file

Ensure the **Game Default Map** is set to the specs of a Meta Quest (1 or 2).

Alternatively, with the HMD plugged in and recognized by your computer, use **LAUNCH**.



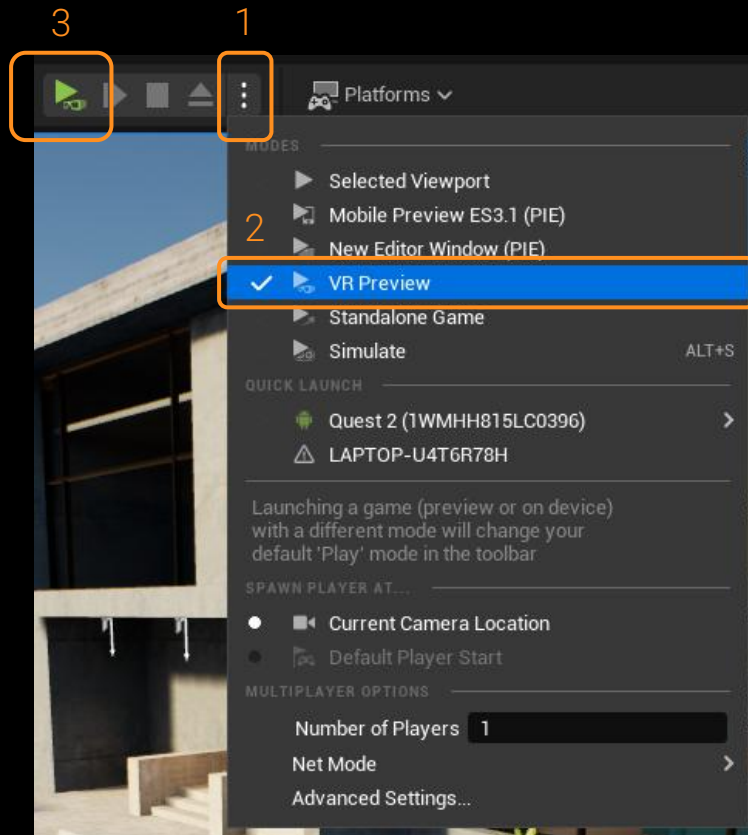
Images courtesy of Cameron Swanson

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Tips & Tricks!

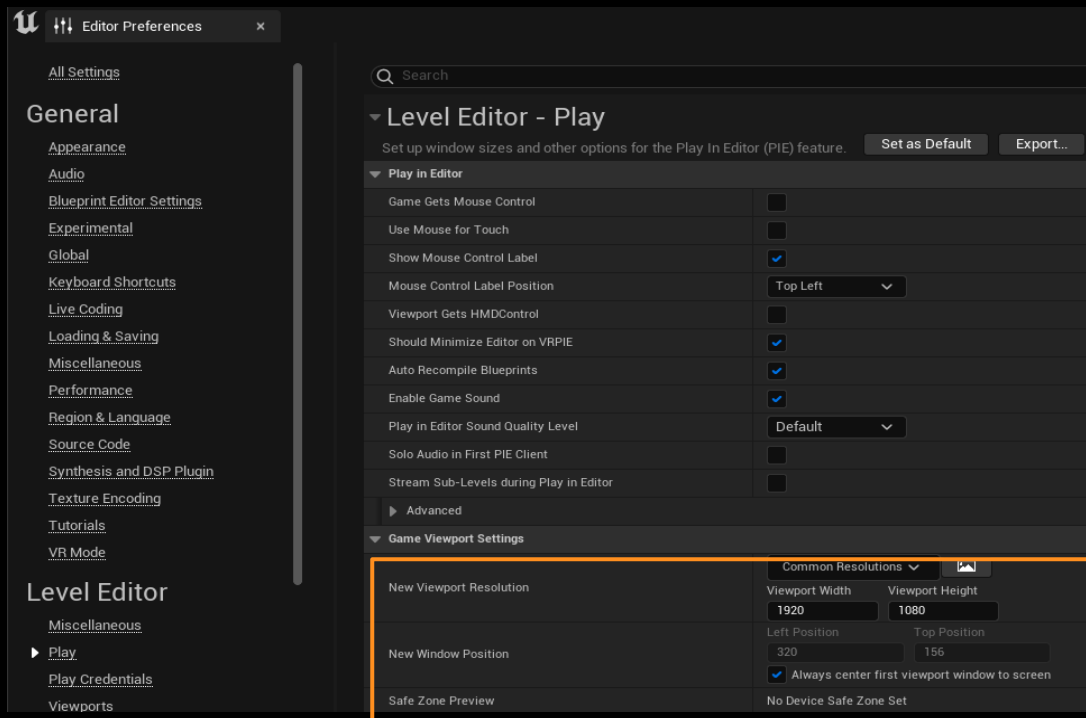
VR - Best Practices

1. Avoid textures if you do not need them.
2. Avoid casting shadows if you do not need them.
3. Minimize dynamic light.
4. Minimize translucency.
5. Instance in visible batches. If one element of your instanced group is visible, the whole group will be drawn.
6. Make LODs for everything above 1000 triangles.
7. Keep material complexity and number of materials per object low.
8. Bake out lighting for anything that you can get away with.
9. Avoid large geometry that can encompass the player.
10. Use precomputed visibility volumes where possible.



VR - Change Preview Window Size

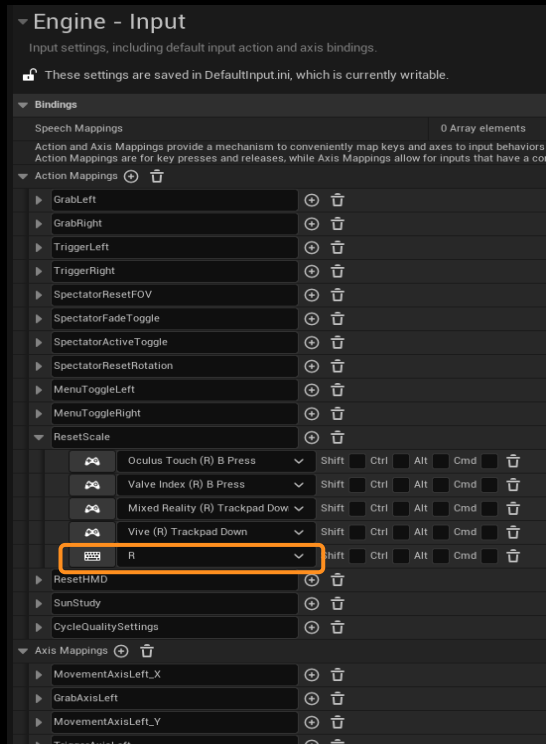
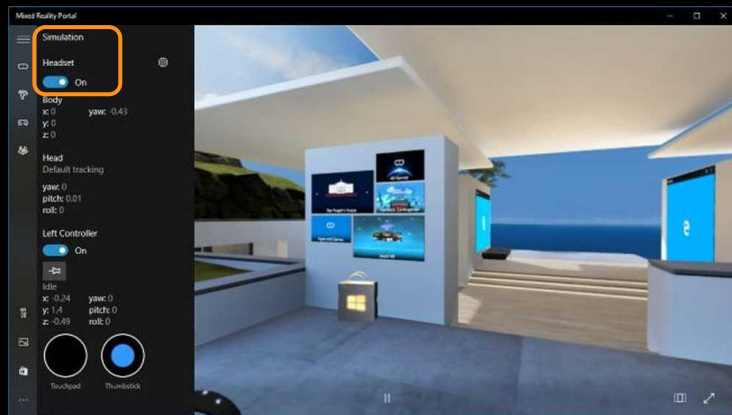
If you want to grow the size of the preview window, look for **Viewport Resolution** in the Editor Preferences:



Intro to Virtual Reality

Can you Simulate VR without an HMD?

- Windows Mixed Reality Portal has a simulation mode.
- Events and Input keys can also be added to simulate without an HMD



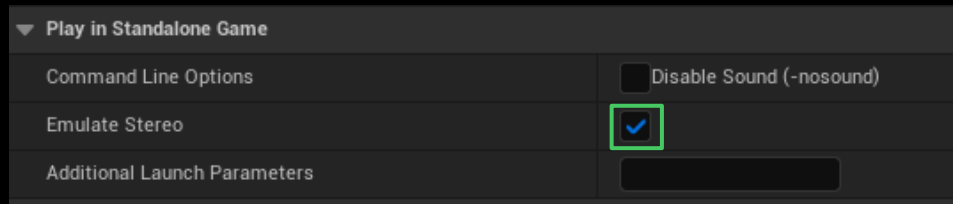
Can you Simulate VR without an HMD?

- Create a non-VR experience
- Create quick links or fast forward features
- Simulate the headset on Desktop

Instructions:

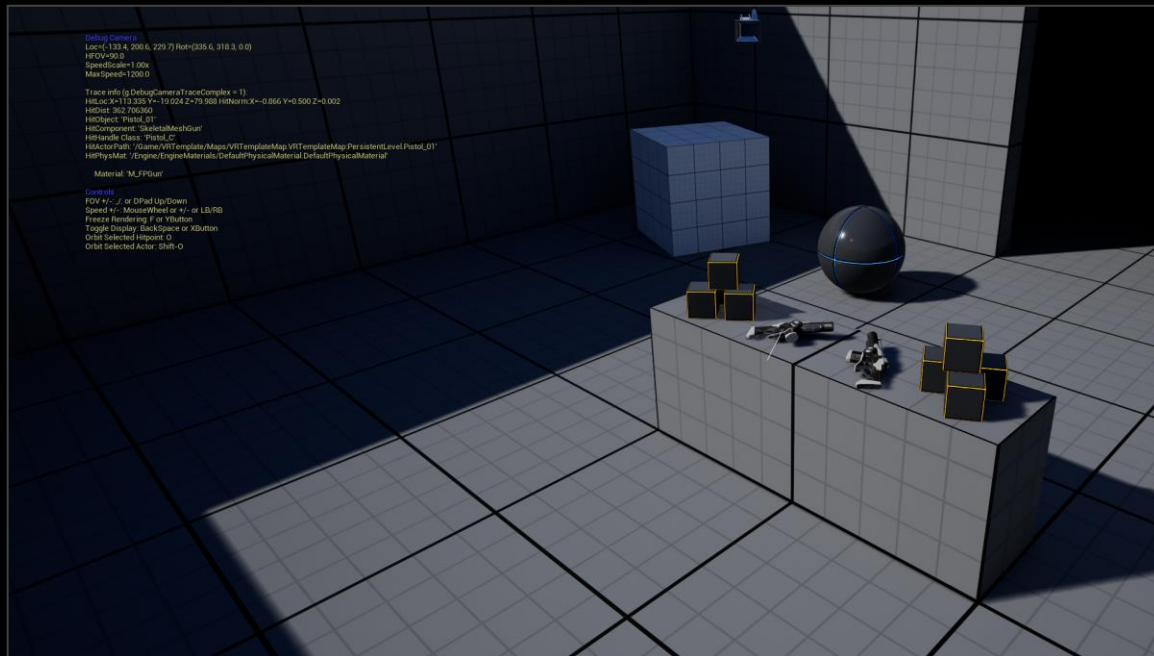
- 1) Launch game or .uproject path with `-game -emulatestereo -res=2160x1200`
- 2) Ensure vsync is off with `r.vsync 0` in the console
- 3) Update screen percentage to emulate oversampling, such as `r.screenpercentage 137`

Alternatively, from Editor you can Play Standalone, using Editor Preferences / Emulate Stereo



Can you Simulate VR without an HMD?

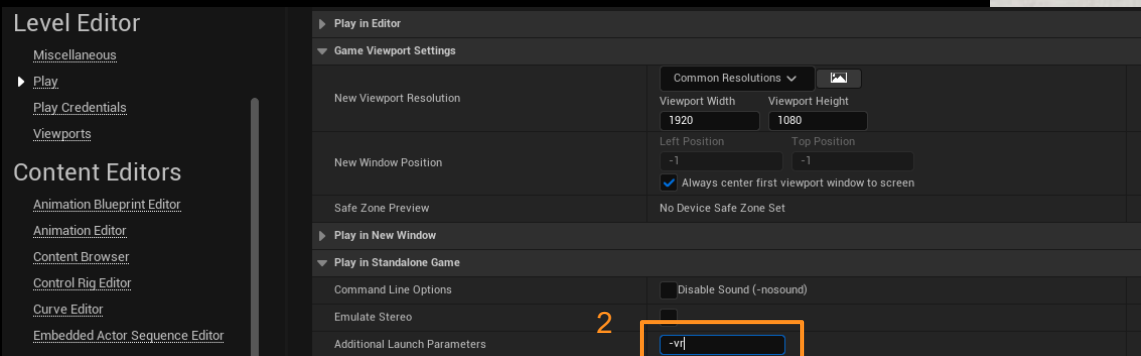
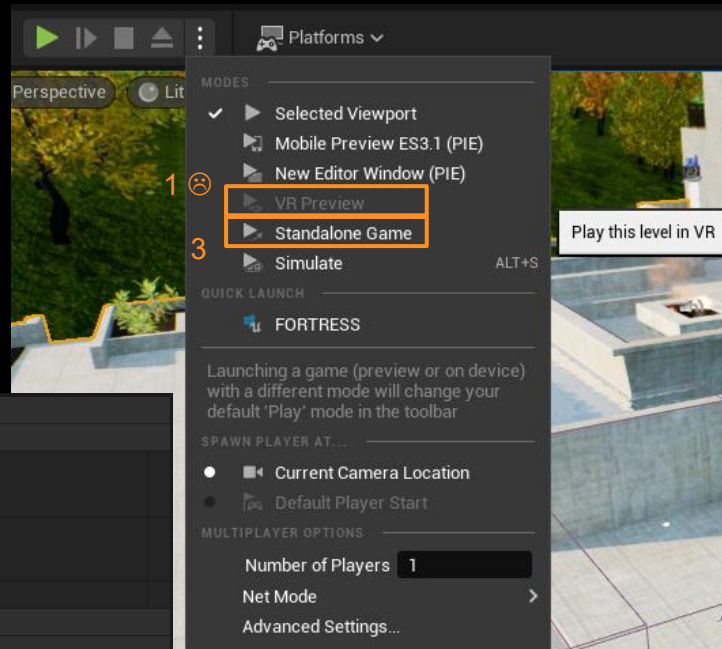
You can also fly around in any development VR experience by typing the cvar `ToggleDebugCamera` or by pressing the **;** (**semicolon**) key.



Can you connect an HMD after entering Editor?

As of 5.2, yes! As long as your OpenXR runtime hasn't changed.

And note if VR Preview is greyed out, after connecting your HMD, instead of restarting the editor, you can also **Play Standalone**... as long as you set up `-vr` as a launch parameter in [Editor Preferences](#).



ADDITIONAL RESOURCES

LEARNING LIBRARY

[Learning | Epic Developer Community \(epicgames.com\)](#)

DOCUMENTATION

- [VR Cheat Sheet](#)
- [OpenXR](#)
- [VR Development](#)

CREDITS

Alex Coulombe



Thank you!